

Accentuation from Nonzero and Multiple Seeds

Uncentered Paths, Calibration Metrics, and the Required RMT Quantities

Theory extension for AccentuationPredRMT

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Abstract

The centered, zero-seed theory reduces own-gradient accentuation to the gain ratio $g = (\beta^*)^\top \hat{\beta} / \|\hat{\beta}\|^2$. Nonzero seeds introduce baseline errors; uncentered steps introduce mean displacement. These are distinct effects. We derive exact conditional-on-fit MSE, coefficient of determination, true-on-predicted regression slope, and Pearson correlation, for one seed, multiple seeds, and seed-dependent target schedules. For independent natural seeds, pooled metrics combine ordinary prediction moments and the same control gain ratio. The additional RMT objects are seed-weighted bilinear and quadratic resolvent forms; finite-fit averages additionally require their joint fluctuations with the gain ratio.

1 Scope, sources, and conventions

Notation follows the main manuscript `accentuation_rmt_main_iclr2026.tex` and its `math_commands.tex` in `AccentuationPredictionRMTDissociation`. This note extends the centered- α derivation in `~/Writings/AccentuationPredictionRMTDissociation/linear_reg_weight_deviation.tex`, and the repository notes `notes/dissociation_prediction_control_feature_space.tex` and `notes/accentuation_varR_theory.tex`. The identities below are direct algebraic derivations; the RMT substitutions are explicitly separated from them. Training remains the centered linear ridge problem of those sources. A nonzero evaluation seed does not require noncentered training data.

Condition on one fitted pixel weight $\hat{\beta}$ (or $\hat{\beta} = F^\top \hat{w}$ for fixed linear features). Allow constant intercepts for completeness:

$$P(\mathbf{x}) = \hat{c} + \hat{\beta}^\top \mathbf{x}, \quad Y(\mathbf{x}) = c_\star + (\beta^*)^\top \mathbf{x}, \quad \mathbf{e} = \hat{\beta} - \beta^*, \quad \delta c = \hat{c} - c_\star.$$

The existing centered theory sets both intercepts to zero. Define

$$\mathbf{x} = \mathbf{x}_0 + \alpha \hat{\beta}, \quad D = \hat{\beta}^\top \hat{\beta} > 0, \quad N = (\beta^*)^\top \hat{\beta}, \quad g = N/D, \quad L = D - N.$$

Here α is the accentuation step, *not* the software ridge penalty; the latter is denoted $\alpha_{\text{Ridge}} = n\lambda$. All evaluation expectations condition on the fit unless explicitly subscripted by “fit.” Teacher responses are noiseless. Slope always means regression of Y on P with a fitted intercept. Our metrics are

$$E = \mathbb{E}(P - Y)^2, \quad R^2 = 1 - \frac{E}{\text{Var } Y}, \quad \text{slope}_p = \frac{\text{Cov}(P, Y)}{\text{Var } P}, \quad r_p = \frac{\text{Cov}(P, Y)}{\sqrt{\text{Var } P \text{Var } Y}}. \quad (1)$$

Thus R^2 evaluates the original predictions, without post-hoc calibration; it is not r_p^2 . Undefined zero-variance cases are not assigned a value.

Notation aligned with the main manuscript. We retain $\beta^*, \hat{\beta}, \mathbf{x}, \Sigma, \hat{\Sigma}, n, d, \lambda, \kappa, N, D$ and reserve $F, \Sigma_F, \hat{w}, w^*$ for linear features. Here $P(\mathbf{x}) = f_{\hat{\beta}}(\mathbf{x})$ and $Y(\mathbf{x}) = f_*(\mathbf{x})$ abbreviate the student and noiseless teacher responses; p_0, y_0 are their seed values. Intercepts are an optional extension and are zero in the main manuscript. We use μ_p, Σ_p for the evaluation input mean and covariance, and μ_0, Σ_0 for seed moments. The generic loss E is E_{acc} on an accentuation distribution and E_{gen} on natural inputs; likewise R^2 is R_{acc}^2 or R_{gen}^2 . The subscript p in slope_p, r_p identifies the evaluation distribution, with r_p reserved for Pearson correlation. Only for compact extension formulas we set

$$g := N/D = \text{slope}_{acc}^{\text{single path}}, \quad z_{acc} = D/N - 1 = g^{-1} - 1 \quad (N \neq 0).$$

Thus g is a path gain, not Pearson r_p or a pooled regression slope. The new seed-residual symbol A_{seed} is a scalar, not a feature map. All traces in this note are ordinary unnormalized Tr ; the main manuscript's normalized trace is $\text{tr}(T) = \text{Tr}(T)/\dim(T)$.

2 One nonzero seed and an uncentered step

Let $\mathbf{x}_0$ be fixed and set

$$p_0 = P(\mathbf{x}_0), \quad y_0 = Y(\mathbf{x}_0), \quad \delta_0 = p_0 - y_0, \quad \mathbb{E}\alpha = m, \quad \text{Var } \alpha = v > 0.$$

Since $P = p_0 + D\alpha$ and $Y = y_0 + N\alpha$,

$$E = vL^2 + (\delta_0 + mL)^2, \quad R^2 = 1 - \frac{vL^2 + (\delta_0 + mL)^2}{vN^2}. \quad (2)$$

For $N \neq 0$,

$$\boxed{\text{slope}_p = g, \quad a = y_0 - gp_0, \quad r_p = \text{sign}(N),} \quad (3)$$

where a is the regression intercept. The means of the step and the seed change calibration error but not this slope or correlation. Pearson correlation is therefore blind to gain magnitude and baseline bias on a single noiseless linear path. For example, $g = 0.1$ gives $r_p = 1$ but, even with zero baseline and centered steps, $R^2 = -80$. A through-origin fit instead gives

$$\text{slope}_{\text{origin}} = \frac{vDN + (p_0 + mD)(y_0 + mN)}{vD^2 + (p_0 + mD)^2};$$

it should not be confused with the gain ratio.

Nonzero seed is not equivalent to uncentered α . Writing $\alpha = m + \tilde{\alpha}$ replaces the seed by $\mathbf{x}_0 + m\hat{\beta}$, so a mean shift can be absorbed into the component of the seed along $\hat{\beta}$. This is an exact reparameterization of the same generated images: every uncentered step can be absorbed into a shifted seed. A general seed, however, has a perpendicular component that cannot be absorbed into the scalar step. Conversely, a nonzero seed need not change $\mathbb{E}\alpha$. The expression $\delta_0 + mL$ displays both effects explicitly.

Does the shifted seed remain natural? No natural-distribution assumption is needed for the exact path identities. It is an additional experimental assumption, useful when accentuation starts from independently held-out natural images. If $\mathbf{x}_0 \sim \mathcal{N}(0, \Sigma)$ independently of the fit and the step, then, conditional on the fit,

$$\tilde{\mathbf{x}}_0 = \mathbf{x}_0 + m\hat{\beta} \sim \mathcal{N}(m\hat{\beta}, \Sigma).$$

Unless $m\hat{\beta} = 0$, this is not the original natural distribution. Unconditionally the shifted seed is also coupled to training through $\hat{\beta}$ (and through m if the schedule is fit-dependent). It cannot be treated as a fresh independent natural seed in an RMT substitution. For a general natural-image distribution, translation likewise need not preserve the distribution or its support. Thus centering α is always possible algebraically, but imposing both centered steps and independent seeds from the original natural distribution restricts the protocol. The mean-bias term in Eq. (??) is retained in the shifted baseline; it has not disappeared.

For seed-dependent schedules one may instead center conditionally: $\tilde{\alpha} = \alpha - \mathbb{E}[\alpha \mid \mathbf{x}_0, \hat{\beta}]$ and $\tilde{\mathbf{x}}_0 = \mathbf{x}_0 + \mathbb{E}[\alpha \mid \mathbf{x}_0, \hat{\beta}]\hat{\beta}$. Then $\mathbb{E}[\tilde{\alpha} \mid \mathbf{x}_0, \hat{\beta}] = 0$ and hence the residual step is uncorrelated with the transformed seed conditional on the fit, but generally not independent of it. This pushforward of the seed distribution can change both its mean and covariance. The master moment formula below still applies. For natural-image experiments we therefore keep the original natural seed and explicitly retain the schedule mean; for algebraic analysis either parameterization is valid.

Student-response normalization. Let $u = D\alpha$, $\mathbb{E}u = \mu$, and $\text{Var } u = \tau^2$. Choosing $\tau^2 = S = (\beta^*)^\top \Sigma \beta^*$ reproduces the old range convention. Then

$$E = \tau^2(1 - g)^2 + [\delta_0 + (1 - g)\mu]^2, \quad (4)$$

$$R^2 = 1 - \frac{(1 - g)^2}{g^2} - \frac{[\delta_0 + (1 - g)\mu]^2}{g^2\tau^2}. \quad (5)$$

Variance matching alone no longer fixes MSE when $\mu \neq 0$. For increment scoring, $\Delta P = u$, $\Delta Y = gu$, so

$$E_\Delta = (\tau^2 + \mu^2)(1 - g)^2, \quad R_\Delta^2 = 1 - \frac{(\tau^2 + \mu^2)(1 - g)^2}{g^2\tau^2}.$$

Subtracting each path’s own mean instead gives centered loss $\tau^2(1 - g)^2$ and the old $R^2 = 1 - (1 - 1/g)^2$. Absolute-response, increment, and centered-response scoring answer different questions. The previous formulas for arbitrary $\mathbf{x}_0$ require the latter convention, or suitable zero-baseline and zero-mean assumptions.

3 Master formula for multiple seeds and dependent schedules

Let $(\mathbf{x}_0, \alpha)$ have any joint evaluation distribution, with finite second moments conditional on the fitted model. Define

$$\mu_0 = \mathbb{E}\mathbf{x}_0, \quad \Sigma_0 = \text{Cov}(\mathbf{x}_0), \quad m = \mathbb{E}\alpha, \quad v = \text{Var } \alpha, \quad \mathbf{c}_{0\alpha} = \text{Cov}(\mathbf{x}_0, \alpha).$$

Here $\mathbf{c}_{0\alpha}$ is the seed–step covariance. The generated input mean and covariance are exactly

$$\mu_p = \mu_0 + m\hat{\beta}, \quad \Sigma_p = \Sigma_0 + v\hat{\beta}\hat{\beta}^\top + \mathbf{c}_{0\alpha}\hat{\beta}^\top + \hat{\beta}\mathbf{c}_{0\alpha}^\top. \quad (6)$$

Define $V_P = \hat{\beta}^\top \Sigma_p \hat{\beta}$, $V_Y = (\beta^*)^\top \Sigma_p \beta^*$, and $C_{PY} = \hat{\beta}^\top \Sigma_p \beta^*$. Then

$E = \mathbf{e}^\top \Sigma_p \mathbf{e} + (\delta c + \mathbf{e}^\top \mu_p)^2, \quad R^2 = 1 - E/V_Y, \quad \text{slope}_p = C_{PY}/V_P, \quad r_p = C_{PY}/\sqrt{V_P V_Y}. \quad (7)$

Explicitly,

$$V_P = \hat{\beta}^\top \Sigma_0 \hat{\beta} + vD^2 + 2D\hat{\beta}^\top \mathbf{c}_{0\alpha}, \quad (8)$$

$$V_Y = (\beta^*)^\top \Sigma_0 \beta^* + vN^2 + 2N(\beta^*)^\top \mathbf{c}_{0\alpha}, \quad (9)$$

$$C_{PY} = \hat{\beta}^\top \Sigma_0 \beta^* + vDN + N\hat{\beta}^\top \mathbf{c}_{0\alpha} + D(\beta^*)^\top \mathbf{c}_{0\alpha}, \quad (10)$$

$$E = \mathbf{e}^\top \Sigma_0 \mathbf{e} + vL^2 + 2L\mathbf{e}^\top \mathbf{c}_{0\alpha} + (\delta c + \mathbf{e}^\top \mu_0 + mL)^2. \quad (11)$$

This includes adaptive schedules and seed selection; no Gaussian evaluation assumption is needed. However, Σ_0 or $\mathbf{c}_{0\alpha}$ depending on the training fit cannot automatically be treated as deterministic weights in an RMT limit.

Finite seed collection. For seeds $\mathbf{x}_j$ with weights π_j summing to one, and conditional step means m_j and variances v_j , use

$$\mu_0 = \sum_j \pi_j \mathbf{x}_j, \quad \Sigma_0 = \sum_j \pi_j (\mathbf{x}_j - \mu_0)(\mathbf{x}_j - \mu_0)^\top, \quad (12)$$

$$m = \sum_j \pi_j m_j, \quad v = \sum_j \pi_j [v_j + (m_j - m)^2], \quad (13)$$

$$\mathbf{c}_{0\alpha} = \sum_j \pi_j (\mathbf{x}_j - \mu_0)(m_j - m). \quad (14)$$

With empirical step moments and observation-count weights, these formulas are exact for the finite pooled data (use variance divisor equal to the number of observations). Equal seed weighting differs from equal image weighting if paths have different lengths. Pooled R^2 , slope, and correlation are generally not averages of per-seed metrics. All seeds share one fitted weight: adding seeds reduces evaluation sampling error, not training-fit uncertainty.

4 Independent seeds: the prediction–control mixture

Suppose $u = D\alpha$ is independent of the seed conditional on the fit, with mean μ and variance τ^2 . Introduce the seed moments

$$D_0 = \hat{\beta}^\top \Sigma_0 \hat{\beta}, \quad N_0 = \hat{\beta}^\top \Sigma_0 \beta^*, \quad S_0 = (\beta^*)^\top \Sigma_0 \beta^*, \quad \delta_{\text{seed}} = \delta c + \mathbf{e}^\top \mu_0.$$

Equation (??) reduces to

$$E = D_0 - 2N_0 + S_0 + \tau^2(1 - g)^2 + [\delta_{\text{seed}} + (1 - g)\mu]^2, \quad (15)$$

$$R^2 = 1 - \frac{E}{S_0 + g^2\tau^2}, \quad (16)$$

$$\text{slope}_p = \frac{N_0 + g\tau^2}{D_0 + \tau^2}, \quad r_p = \frac{N_0 + g\tau^2}{\sqrt{(D_0 + \tau^2)(S_0 + g^2\tau^2)}}. \quad (17)$$

When $D_0 > 0$, the pooled slope is the variance-weighted average

$$\text{slope}_p = \frac{D_0}{D_0 + \tau^2} \frac{N_0}{D_0} + \frac{\tau^2}{D_0 + \tau^2} g.$$

The normalization $\tau^2 = S$ matches the *within-seed student variance*; the pooled student variance is $D_0 + S$, not S . If instead total student variance must equal S , one needs $\tau^2 = S - D_0 \geq 0$ under this independent protocol.

For centered natural seeds, $\Sigma_0 = \Sigma$, $\mu_0 = 0$, zero intercepts, and centered increments, let $E_{\text{gen}} = D_0 - 2N_0 + S$. Then

$$\boxed{E = E_{\text{gen}} + \tau^2(1 - g)^2, \quad R_{\text{pooled}}^2 = 1 - \frac{E_{\text{gen}} + \tau^2(1 - g)^2}{S + g^2\tau^2}.} \quad (18)$$

At $\tau^2 = S$, this is $R_{\text{pooled}}^2 = 1 - [E_{\text{gen}}/S + (1 - g)^2]/(1 + g^2)$. Thus multiple natural seeds bring ordinary generalization error back into accentuation evaluation. At $\tau^2 = 0$, all metrics are seed-prediction metrics. For fixed $g \neq 0$ and $\mu = 0$, as τ^2 dominates seed variance, slope tends to g , Pearson correlation tends to $\text{sign}(g)$, and R^2 tends to the single-path value. At $g = 0$, seed teacher variance can keep pooled R^2 defined even though per-path R^2 is undefined.

5 A common alternative: absolute student target levels

Suppose the experiment selects a target response q , independently of seeds, and moves until $P(\mathbf{x}) = q$. Then

$$\alpha = (q - p_0)/D, \quad P = q, \quad Y = gq + a_0, \quad a_0 = y_0 - gp_0.$$

This makes the step seed-dependent, even for a centered target distribution. Let $\mathbb{E}q = \nu$, $\text{Var } q = t^2 > 0$, $\mathbb{E}a_0 = \bar{a}$, and $\text{Var } a_0 = v_a$. Direct calculation gives

$$E = (1 - g)^2 t^2 + v_a + [(1 - g)\nu - \bar{a}]^2, \quad (19)$$

$$R^2 = 1 - \frac{E}{g^2 t^2 + v_a}, \quad \text{slope}_p = g, \quad (20)$$

$$r_p = \frac{gt}{\sqrt{g^2 t^2 + v_a}}, \quad v_a = (\beta^* - g\hat{\beta})^\top \Sigma_0 (\beta^* - g\hat{\beta}). \quad (21)$$

Here $c_{0\alpha} = -\Sigma_0 \hat{\beta}/D$. Unlike independent increments, the pooled slope remains g , while Pearson correlation measures the variation of path intercepts as well. Consequently the experimental schedule must be specified before quoting a multi-seed metric formula.

6 A specific protocol: natural seeds and natural response matching

Take independent held-out natural seeds with mean $\mathbf{m}_{\text{nat}}$ and covariance Σ , and condition throughout on one fitted model. Define

$$\begin{aligned} \nu_P &= \hat{c} + \hat{\beta}^\top \mathbf{m}_{\text{nat}}, & \nu_Y &= c_\star + (\beta^\star)^\top \mathbf{m}_{\text{nat}}, & \delta &= \nu_P - \nu_Y, \\ D_{\text{gen}} &= \hat{\beta}^\top \Sigma \hat{\beta}, & N_{\text{gen}} &= \hat{\beta}^\top \Sigma \beta^\star, & S &= (\beta^\star)^\top \Sigma \beta^\star > 0, & g &= N/D. \end{aligned}$$

The reference distribution is the *noiseless natural teacher response*, with mean ν_Y and variance S . We first require the generated *student absolute response* to match these moments. This is the nonzero-seed analogue of the original student-range normalization; it does not assume that the generated teacher response is already calibrated. Matching two moments implies full distributional equality only in a Gaussian setting, or when we explicitly sample target responses from the reference law.

6.1 Match the natural response range separately from each seed

For every seed, draw an absolute target q independently from the natural teacher response distribution and set

$$\alpha = \frac{q - P(\mathbf{x}_0)}{D}. \quad (22)$$

Then $P(\mathbf{x}_0 + \alpha\hat{\beta}) = q$ exactly, so the entire student response distribution matches the reference, conditionally on *each* seed. In particular,

$$\mathbb{E}[\alpha \mid \mathbf{x}_0, \hat{\beta}] = \frac{\nu_Y - p_0}{D}, \quad \text{Var}(\alpha \mid \mathbf{x}_0, \hat{\beta}) = \frac{S}{D^2}. \quad (23)$$

For Gaussian natural inputs these conditional steps are Gaussian. For a non-Gaussian response reference, use its actual quantiles or samples for q ; mean and variance alone do not specify its law.

A seed with a high predicted baseline needs a negative mean step; one with a low predicted baseline needs a positive mean step. Even if the global natural response is centered, the step is generally not centered for an individual seed. Averaging over natural seeds gives, by total expectation and variance,

$$\mathbb{E}\alpha = -\frac{\delta}{D}, \quad \text{Var} \alpha = \frac{S + D_{gen}}{D^2}, \quad \text{Cov}(\mathbf{x}_0, \alpha) = -\frac{\Sigma\hat{\beta}}{D}. \quad (24)$$

The marginal step variance is larger than the per-seed variance: it includes the variation of the baseline-canceling conditional means. For centered zero-intercept models, $\mathbb{E}\alpha = 0$ marginally but $\mathbb{E}[\alpha \mid \mathbf{x}_0] = -p_0/D$ is usually nonzero. Ignoring this seed-step anticorrelation gives an incorrect pooled variance. Indeed,

$$\text{Var} P_{acc} = D_{gen} + D^2 \text{Var} \alpha + 2D \text{Cov}(p_0, \alpha) = D_{gen} + (S + D_{gen}) - 2D_{gen} = S.$$

Conditionally centering the step produces the shifted seed

$$\tilde{\mathbf{x}}_0 = \mathbf{x}_0 + \frac{\nu_Y - p_0}{D}\hat{\beta}, \quad P(\tilde{\mathbf{x}}_0) = \nu_Y,$$

with mean $\mathbf{m}_{nat} - \delta\hat{\beta}/D$ and covariance $J\Sigma J^\top$, where $J = I - \hat{\beta}\hat{\beta}^\top/D$. Thus the shifted seeds lie on a common student-response hyperplane and are not independent samples from the original natural distribution. The original seeds are natural; the recentered mathematical seeds need not be.

6.2 Metrics and the remaining RMT quantity

Let $a_0 = y_0 - gp_0$ and

$$A_{seed} = \text{Var}(a_0) = (\beta^* - g\hat{\beta})^\top \Sigma (\beta^* - g\hat{\beta}) = S - 2gN_{gen} + g^2D_{gen}. \quad (25)$$

The teacher response on the generated images is $Y_{acc} = gq + a_0$. Since q and a_0 are independent,

$$\mathbb{E}Y_{acc} = \nu_Y - g\delta, \quad \text{Var} Y_{acc} = g^2S + A_{seed}, \quad (26)$$

$$E = (1 - g)^2S + A_{seed} + g^2\delta^2, \quad (27)$$

$$R^2 = 1 - \frac{(1 - g)^2S + A_{seed} + g^2\delta^2}{g^2S + A_{seed}}, \quad (28)$$

$$\text{slope}_p = g, \quad r_p = \frac{g\sqrt{S}}{\sqrt{g^2S + A_{seed}}}. \quad (29)$$

The denominators must be nonzero. In the centered case, the compact dimensionless form is

$$\frac{E}{S} = (1 - g)^2 + \frac{A_{\text{seed}}}{S}, \quad R^2 = 1 - \frac{(1 - g)^2 + A_{\text{seed}}/S}{g^2 + A_{\text{seed}}/S}, \quad r_p = \frac{g}{\sqrt{g^2 + A_{\text{seed}}/S}}.$$

Thus natural-response matching leaves two distinct sources of error: path gain miscalibration and variation of the teacher response across equal-student-response seed locations. Even though student responses have the correct marginal law, the teacher law generally does not match it. The pooled slope is still g , but Pearson correlation is no longer forced to be ± 1 . The additional RMT summary is A_{seed}/S , constructed from the existing natural-prediction moments $N_{\text{gen}}, D_{\text{gen}}$ and the control ratio g . A leading concentrated-fit DE uses

$$A_{\text{seed}}^{\text{DE}} = S - 2g_{\text{DE}}N_{\text{gen},\text{DE}} + g_{\text{DE}}^2 D_{\text{gen},\text{DE}}.$$

Its fit average instead requires $\mathbb{E}[gN_{\text{gen}}]$ and $\mathbb{E}[g^2 D_{\text{gen}}]$, so joint fluctuations matter when these quantities do not concentrate. As a check, if $\hat{\beta} = k\beta^*$ with $k \neq 0$ and zero intercepts, $g = 1/k$ and $A_{\text{seed}} = 0$: multi-seed correlation is still $\text{sign}(k)$, while gain miscalibration remains. At $\hat{\beta} = \beta^*$, all metrics are perfect even though the protocol can move images substantially.

6.3 Pooled matching alone does not identify a unique schedule

Suppose we only require the *pooled* student responses to have mean ν_Y and variance S . Put $c_{p\alpha} = \text{Cov}(p_0, \alpha)$. Then

$$\mathbb{E}\alpha = -\delta/D, \quad D^2 \text{Var } \alpha + 2Dc_{p\alpha} = S - D_{\text{gen}}. \quad (30)$$

Without specifying dependence on the seed, the variance is not uniquely identified. Under *independent steps*, $c_{p\alpha} = 0$, hence

$$\text{Var } \alpha = (S - D_{\text{gen}})/D^2,$$

which is feasible only for $D_{\text{gen}} \leq S$. In the centered case this adds only the missing student variance; if $D_{\text{gen}} = S$, it forces no motion. If $D_{\text{gen}} > S$, independent steps cannot reduce response variance. Independent Gaussian steps give exact Gaussian response matching when natural seeds are Gaussian, but only pooled matching, not matching separately from each seed.

More generally, couple a target q with the seed while keeping its desired marginal distribution, and let $c = \text{Cov}(q, p_0)$. Equation (??) gives

$$\text{Var } \alpha = \frac{S + D_{\text{gen}} - 2c}{D^2}, \quad |c| \leq \sqrt{SD_{\text{gen}}}.$$

The independent-target protocol above chooses $c = 0$. For Gaussian seeds with $D_{\text{gen}} > 0$, the deterministic choice $q = \nu_Y + \sqrt{S/D_{\text{gen}}}(p_0 - \nu_P)$ gives the same Gaussian pooled reference law with $c = \sqrt{SD_{\text{gen}}}$ and the smallest possible step variance $(\sqrt{S} - \sqrt{D_{\text{gen}}})^2/D^2$. It yields one point per seed rather than a full response range from every seed. The metrics in the previous subsection assume independent targets and must not be reused for this correlated-target protocol.

6.4 Which response is being matched?

Matching generated student responses to *natural student* responses would use (ν_P, D_{gen}) instead of (ν_Y, S) . Independent-step pooled matching then forces $\alpha = 0$ almost surely, whereas independent absolute targets still allow motion, with conditional variance D_{gen}/D^2 and marginal variance $2D_{\text{gen}}/D^2$.

If the requirement instead concerns the *true teacher* response after accentuation, its moments obey

$$\mathbb{E}Y_{\text{acc}} = \nu_Y + N\mathbb{E}\alpha, \quad \text{Var } Y_{\text{acc}} = S + N^2 \text{Var } \alpha + 2N \text{Cov}(y_0, \alpha).$$

For $N \neq 0$, matching the natural teacher moments with seed-independent steps forces $\mathbb{E}\alpha = 0$ and $\text{Var } \alpha = 0$. Nontrivial matching therefore needs seed-dependent steps, for example the oracle schedule $\alpha = (q - y_0)/N$ with an independent natural teacher target q . This is not the student-controlled protocol and requires teacher information. At $N = 0$, every step leaves teacher responses unchanged, so teacher-distribution matching imposes no constraint on motion.

Finally, exact moment matching can conflict with purely one-sided accentuation ($\alpha \geq 0$). In the centered, equal-mean case it requires $\mathbb{E}\alpha = 0$, which under that constraint forces $\alpha = 0$ almost surely. A nontrivial matched-range experiment generally includes both accentuation and de-accentuation, or deliberately uses a shifted reference distribution.

7 The key RMT quantities

7.1 Which scalars are sufficient?

For independent increments, the conditional metrics depend on

$$\boxed{(N, D, N_0, D_0, \ell_0 = \hat{\beta}^\top \mu_0),} \quad (31)$$

together with deterministic teacher moments, intercepts, and schedule moments. A single fixed seed has $\Sigma_0 = 0$, so the new object is $\ell_0 = \hat{\beta}^\top \mathbf{x}_0$, or equivalently δ_0 . For centered natural seeds, N_0, D_0 are exactly the existing $N_{\text{gen}}, D_{\text{gen}}$; no new leading-order scalar theory is needed. For general seeds, the new geometry is Σ_0 , not necessarily Σ or I . For dependent schedules add $\hat{\beta}^\top \mathbf{c}_{0\alpha}$ and $(\beta^*)^\top \mathbf{c}_{0\alpha}$; when the schedule depends on the fit, express these through the actual protocol before applying a DE.

7.2 Ridge resolvents and leading deterministic equivalents

For direct pixel ridge let

$$\hat{\Sigma} = X^\top X/n, \quad Q = (\hat{\Sigma} + \lambda I)^{-1}, \quad \hat{\beta} = (I - \lambda Q)\beta^* + \frac{\sigma}{n} Q X^\top \eta.$$

Conditional on X , write $\hat{\beta} = \mathbf{m}_X + \boldsymbol{\xi}$ with $\mathbf{m}_X = (I - \lambda Q)\beta^*$ and $\Omega_X = (\sigma^2/n)Q\hat{\Sigma}Q$. For deterministic Σ_0 and $\mathbf{a}$, the exact conditional identities are

$$\mathbb{E}_\eta[\mathbf{a}^\top \hat{\beta} \mid X] = \mathbf{a}^\top \mathbf{m}_X, \quad \mathbb{E}_\eta[\hat{\beta}^\top \Sigma_0 \hat{\beta} \mid X] = \mathbf{m}_X^\top \Sigma_0 \mathbf{m}_X + \text{Tr}(\Sigma_0 \Omega_X).$$

Thus the required random-matrix objects are one-resolvent bilinear forms $\mathbf{a}^\top Q \beta^*$, weighted two-resolvent forms $(\beta^*)^\top Q \Sigma_0 Q \beta^*$, and traces $\text{Tr}(\Sigma_0 Q \hat{\Sigma} Q)$. Replacing Q independently in each factor of $Q \Sigma_0 Q$ loses design variance.

Using the same leading aggregate DE as the existing feature note, set

$$\kappa - \lambda = \frac{\kappa}{n} \text{Tr}[\Sigma(\Sigma + \kappa I)^{-1}], \quad (32)$$

$$H = (\Sigma + \kappa I)^{-1}, \quad \tilde{\beta}_\kappa = \Sigma H \beta^*, \quad (33)$$

$$v_0 = \frac{\kappa^2 (\beta^*)^\top H \Sigma H \beta^* + \sigma^2}{n - \text{Tr}(\Sigma^2 H^2)}, \quad V_{\beta, DE} = v_0 H \Sigma H. \quad (34)$$

In the main manuscript’s spectral notation,

$$B_{a,b}(\kappa) = \sum_k \frac{s_k^a b_k^2}{(s_k + \kappa)^b}, \quad \text{df}_{a,b}(\kappa) = \sum_k \frac{s_k^a}{(s_k + \kappa)^b}, \quad \text{df}_2 = \text{df}_{2,2}.$$

Hence $v_0 = (\kappa^2 B_{1,2} + \sigma^2)/(n - \text{df}_2) = (E_{\text{gen},DE} + \sigma^2)/n$. The manuscript’s training-fit means $\mu_N = \mathbb{E}_{X,\eta} N$ and $\mu_D = \mathbb{E}_{X,\eta} D$ have leading equivalents $N_{\text{DE}} = B_{1,1}$ and $D_{\text{DE}} = B_{2,2} + v_0 \text{df}_{1,2}$, respectively; these DEs are not exact finite-sample means.

Then the aggregate substitutions are

$$N_{\text{DE}} = (\beta^\star)^\top \tilde{\beta}_\kappa, \quad D_{\text{DE}} = \|\tilde{\beta}_\kappa\|^2 + \text{Tr } V_{\beta,DE}, \quad (35)$$

$$N_{0,DE} = \tilde{\beta}_\kappa^\top \Sigma_0 \beta^\star, \quad D_{0,DE} = \tilde{\beta}_\kappa^\top \Sigma_0 \tilde{\beta}_\kappa + \text{Tr}(\Sigma_0 V_{\beta,DE}), \quad (36)$$

$$\ell_{0,DE} = \mu_0^\top \tilde{\beta}_\kappa, \quad g_{\text{DE}} = N_{\text{DE}}/D_{\text{DE}}. \quad (37)$$

These are leading aggregate equivalents under the same fixed-geometry, boundedness, and concentration assumptions as the source note. $V_{\beta,DE}$ is an aggregate second-moment prescription, not an exact finite-dimensional covariance or a full joint CLT. In particular, for a fixed finite number of seeds, baseline projections need not concentrate on the scale relevant to their squared errors. When they do not, retain the exact conditional Gaussian moments above or a justified joint fluctuation approximation.

In a common eigenbasis of Σ_0 and Σ , with eigenvalues $s_{0,k}$ and s_k , and teacher coordinates b_k , the new quadratic form reads

$$D_{0,DE} = \sum_k s_{0,k} \left[\left(\frac{s_k b_k}{s_k + \kappa} \right)^2 + v_0 \frac{s_k}{(s_k + \kappa)^2} \right].$$

This shows exactly how seed variability weights the same spectral quantities. Noncommuting Σ_0 requires the matrix form, not just its diagonal spectrum.

7.3 Linear features

Use the existing notation $\Sigma_F = F \Sigma F^\top$, $G = F F^\top$, $\mathbf{q} = F \beta^\star$, and population feature coefficient $w^\star$. With $m_{w,DE}$ and $V_{w,DE}$ from the feature-space DE, replace

$$\tilde{\beta}_\kappa = F^\top m_{w,DE}, \quad V_{\beta,DE} = F^\top V_{w,DE} F.$$

Control uses G ; seed prediction adds $\Sigma_{0,F} = F \Sigma_0 F^\top$ and $\mathbf{q}_0 = F \Sigma_0 \beta^\star$. In particular,

$$N_{0,DE} = m_{w,DE}^\top \mathbf{q}_0, \quad D_{0,DE} = m_{w,DE}^\top \Sigma_{0,F} m_{w,DE} + \text{Tr}(\Sigma_{0,F} V_{w,DE}).$$

Thus training covariance, control geometry, and seed covariance are three potentially different matrices. The teacher residual outside the feature span is already included by retaining the full teacher S_0 and $\mathbf{q}_0$.

7.4 Averaging over fitted models needs joint fluctuations

Even the one-seed normalized MSE is not determined by $\mathbb{E}g$ alone. For fixed μ, τ^2 , exactly

$$\mathbb{E}_{\text{fit}} E = \tau^2 \{ (1 - \bar{g})^2 + \text{Var}_{\text{fit}} g \} + [\bar{\delta}_0 + \mu(1 - \bar{g})]^2 \quad (38)$$

$$+ \text{Var}_{\text{fit}} \delta_0 + \mu^2 \text{Var}_{\text{fit}} g - 2\mu \text{Cov}_{\text{fit}}(\delta_0, g). \quad (39)$$

Thus the new fluctuation object is the *joint* law of baseline error and control gain, not only $\text{Var}(g)$. Writing $\mathbf{j}_g = (\beta^* - 2g\hat{\beta})/D$ gives the local differential $dg = \mathbf{j}_g^\top d\hat{\beta}$. At fixed X , a first-order Gaussian delta calculation around $\mathbf{m}_X$ (where its norm is nonzero) gives

$$\text{Var}_\eta(\delta_0 \mid X) = \mathbf{x}_0^\top \Omega_X \mathbf{x}_0, \quad \text{Cov}_\eta(\delta_0, g \mid X) \simeq \mathbf{x}_0^\top \Omega_X \mathbf{j}_g \big|_{\hat{\beta}=\mathbf{m}_X}.$$

This assumes fixed intercepts and small conditional fluctuations. Full unconditional covariance also contains variation of conditional means over X , as in the existing variance note. Pooled ratio metrics require the joint law of Eq. (??); CV additionally randomizes λ . Neither ratios of means nor a Gaussian fitted-weight approximation are exact expectations of nonlinear metrics. Near $N = 0$ or a small true-response variance, report distributions/quantiles and assess denominator concentration.

8 Practical reporting and limits

For each experiment specify: absolute responses versus increments versus within-path centering; independent increments versus absolute targets; step mean and variance; seed/observation weights; and pooled versus per-seed metrics. Report MSE and uncalibrated R^2 alongside slope and Pearson r_p . A high Pearson r_p on a single linear path establishes monotonicity only. Pooling seed baselines can also yield high correlation even when the control gain is poor; within-seed gains make that failure visible.

If independent evaluation noise of variance σ_{test}^2 is added to each teacher response, add it to both MSE and V_Y , leaving C_{PY} and V_P unchanged. Slope is unchanged, correlation is attenuated, and the noiseless ± 1 single-path result no longer holds. Subtracting a noisy seed response creates shared noise across increments and needs its covariance explicitly included.

All exact results assume a common, constant gradient $\hat{\beta}$. Nonlinear networks, clipping, or image constraints generally make gains seed- and step-dependent. A local linearization can apply per seed, but a pooled analysis then needs the distribution of those local gains and their covariance with baselines; the single ratio N/D is no longer sufficient.